

Centroid and Moment of Inertia

Centroid:-

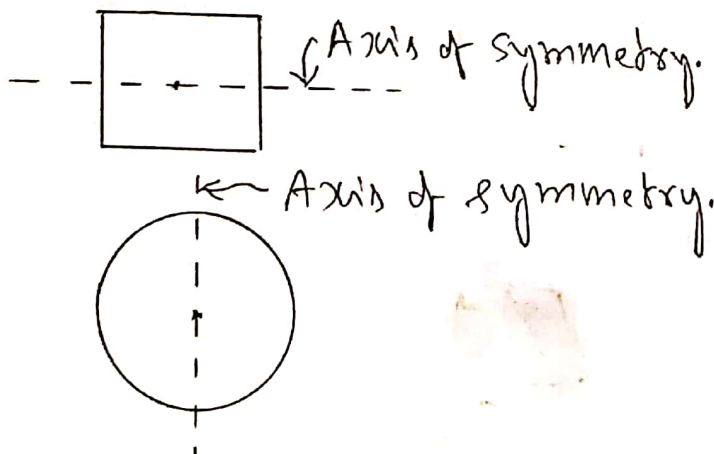
Definition of Centroid:

"Centroid is defined as the point where whole area of the body is assumed to be concentrated."

Center of gravity:- "Center of gravity may be defined as the point where the whole mass of the body assumed to be concentrated."

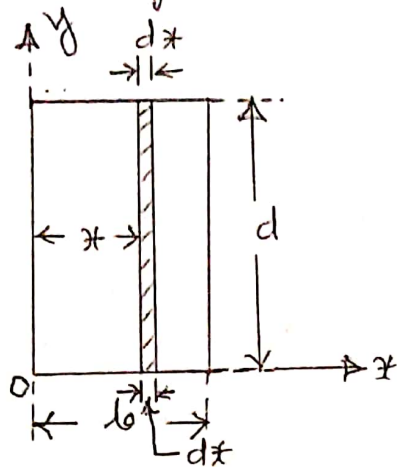
Axis of symmetry:- If the given area is equally distributed on either side of the axis, the axis is called axis of symmetry.

Eg:



Centroids of Regular figures:

i) Rectangle:- (Centroid)



Let the width of rectangle is 'b' and depth is 'd'. Let us consider an elementary strip of width 'dx' at a distance 'x' from y-axis.

Area of elementary strip = $d \times dx$.

Moment of elementary strip about y-axis
ie = $d \times dx \times x$

Moment of whole area about y-axis,

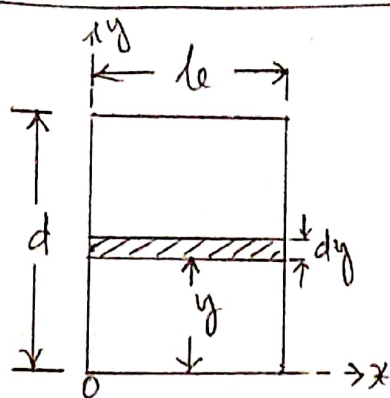
$$\begin{aligned} \text{ie} &= \int_0^b d \times dx \times x \\ &= d \left[\frac{x^2}{2} \right]_0^b = d \frac{b^2}{2} \end{aligned}$$

Total area of rectangle = $b \times d$

$\therefore$ Centroidal distance from y-axis,

$$\begin{aligned} \text{ie } \bar{x} &= \frac{\text{Moment of Total area}}{\text{Total area}} \\ &= \frac{\frac{b^2}{2} d}{b \times d} = \frac{b}{2} \end{aligned}$$

$$\boxed{\therefore \bar{x} = \frac{b}{2}}$$



Let us consider a elementary strip of thickness 'dy' at a distance 'y' from x-axis.

Area of strip = $b \times dy$.

Moment of strip about x-axis,

$$ie = b \times dy \times y.$$

$\therefore$ Moment of whole area about x-axis.

$$ie = \int_0^d b \times dy \times y$$

$$= b \left[\frac{y^2}{2} \right]_0^d$$

$$= \frac{b \times d^2}{2}.$$

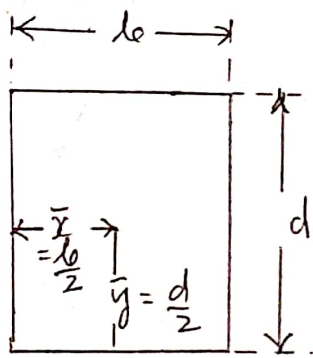
Area of rectangle = $b \times d$.

$\therefore$ Centroid distance of rectangle from x-axis, ~~xxx~~

$$ie = \bar{y} = \frac{\text{Moment of whole area}}{\text{Area of rectangle}}$$

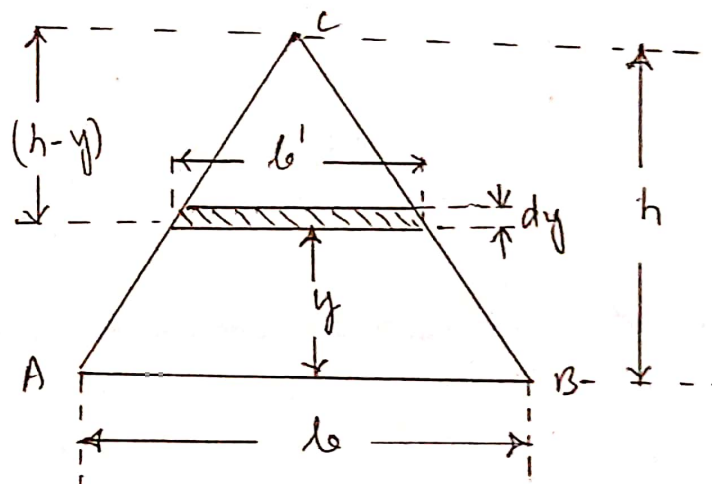
$$= \frac{\frac{b d^2}{2}}{b \times d} = \frac{d}{2}$$

$$\boxed{\bar{y} = \frac{d}{2}}$$



ii) Centroid of Triangle:-

Let us consider a triangle ABC of base ' b ' and height ' h '.



Let us consider an elementary strip of thickness ' dy ' at a distance ' y ' from base AB.

From A^k above, Area of elementary strip = $b' \times dy$.

$$\frac{b}{b'} = \frac{h}{(h-y)}$$

$$b' = \frac{b(h-y)}{h}$$

$$\therefore \text{Area of elementary strip} = \frac{b(h-y)}{h} \times dy$$

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Moment of elementary strip about base
 $AB = \frac{b(h-y)}{h} \times dy \times y.$

$$= \frac{b \times h \left(1 - \frac{y}{h}\right)}{h} dy \times y$$

$$= b \left(1 - \frac{y}{h}\right) y \times dy$$

$$= b \times y \times dy - \frac{b \times y^2 dy}{h}.$$

$\therefore$ Moment of whole area about base

$$AB = \int_0^h b \times y \times dy - \int_0^h \frac{b y^2 dy}{h}$$

$$= b \left[\frac{y^2}{2} \right]_0^h - \frac{b}{h} \left[\frac{y^3}{3} \right]_0^h$$

$$= b \frac{h^2}{2} - \frac{b h^3}{3h}$$

$$= \frac{b h^2}{2} - \frac{b h^2}{3}$$

$$= \frac{b h^2}{6}$$

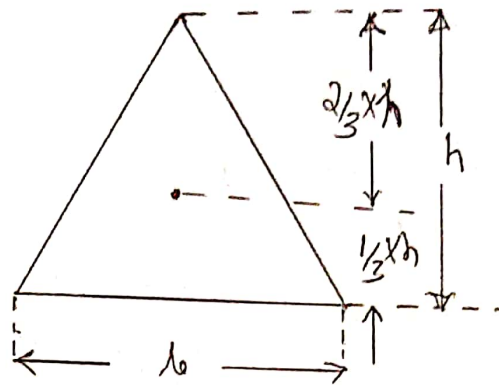
$$\text{Area of } \Delta b = \frac{1}{2} \times b \times h$$

Distance of centroid of Δb from base AB

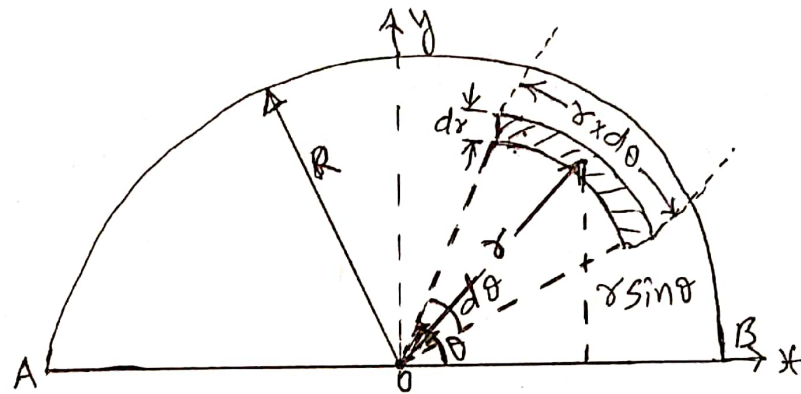
$$ie = \frac{\text{Moment of whole area about base}}{\text{Area of triangle}}$$

$$= \frac{\frac{b h^2}{6}}{\frac{b h}{2}} = \frac{b h^2}{6} \times \frac{2}{b h} = \frac{2h}{6} = \frac{h}{3}$$

∴ Centroid of a Δ is at a distance of $\frac{h}{3}$ from base and $\frac{2h}{3}$ from apex.



iii) Centroid of Semicircle :-



Let us consider a semicircle of radius 'R'. Consider an elementary strip of thickness 'dr' at a radius 'r' from 'O'. Let ~~let~~ $d\theta$ be the angle subtended by elementary strip at 'O'. Let θ be the angle of radius 'r' with respect to x-axis.

Let length of elementary strip = $r \times d\theta$.

Area of elementary strip = $r \times d\theta \times dr$.

Moment of elementary strip about base

$$AB = r \times d\theta \times dr \times r \sin \theta = r^2 \times dr \sin \theta d\theta$$

Moment of whole area about base,

$$\text{i.e.} = \int_0^R \int_0^\pi r^2 dr \times \sin \theta d\theta = \int_0^R r^2 dr \int_0^\pi \sin \theta d\theta$$

$$= \left[\frac{y^3}{3} \right]_0^R [-\cos \theta]_0^\pi$$

$$= \frac{R^3}{3} [1+1] = \frac{2R^3}{3}$$

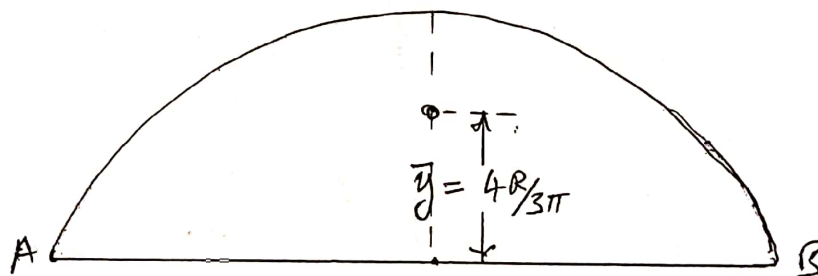
$$\text{Area of semicircle} = \frac{\pi R^2}{2}$$

$\therefore$ Distance of centroid from base,

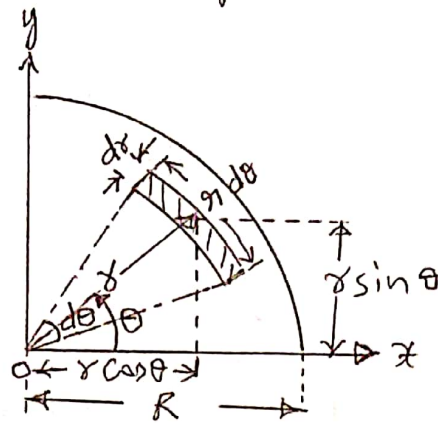
$$AB = \frac{\text{Moment of whole area about base}}{\text{Area of semicircle}}$$

$$= \frac{\frac{2\pi R^3}{3}}{\frac{\pi R^2}{2}} = \frac{2R^3}{3} \times \frac{2}{\pi R^2}$$

$$= \frac{2R^3}{3} \times \frac{2}{\pi R^2} = \frac{4R}{3\pi}$$



iii) Centroid of Quarter Circle



Consider a quarter circle of radius 'R' as shown in fig. The details of elementary strip are also shown in fig above.

Area of the element $= r d\theta \times dr$

Moment of elementary strip about OX-axis
 $= r d\theta \times dr \times r \sin \theta$

$$= r^2 dr \sin \theta d\theta.$$

$\therefore$ Moment of whole area about the base OX

$$= \int_0^R \int_0^{\pi/2} r^2 dr \sin \theta d\theta$$

$$= \int_0^R r^2 dr \int_0^{\pi/2} \sin \theta d\theta$$

$$= \left[\frac{r^3}{3} \right]_0^R \times \left[-\cos \theta \right]_0^{\pi/2}$$

$$= \frac{R^3}{3} \times [0 - (-1)] = \frac{R^3}{3}.$$

$$\text{Area of quarter circle} = \frac{\pi R^2}{4}$$

$$\therefore \bar{y} = \frac{\text{Moment of area about OX axis}}{\text{area of quarter circle}}$$

$$= \frac{R^3/3}{\pi R^2/4} = \frac{R^3}{3} \times \frac{4}{\pi R^2} = \frac{4R}{3\pi}$$

Similarly.

Moment of elementary area about OY-axis

$$= x \, d\theta \times dy \times x \cos \theta$$

$$= x^2 dy \cos \theta \, d\theta$$

Moment of whole area about OY axis

$$= \int_0^R x^2 dy \int_0^{\pi/2} \cos \theta \, d\theta$$

$$= \left[\frac{x^3}{3} \right]_0^R \times \left[\sin \theta \right]_0^{\pi/2}$$

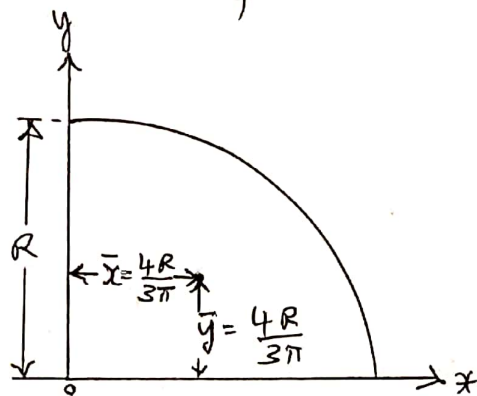
$$= \frac{R^3}{3} \times (1 - 0)$$

$$= \frac{R^3}{3}$$

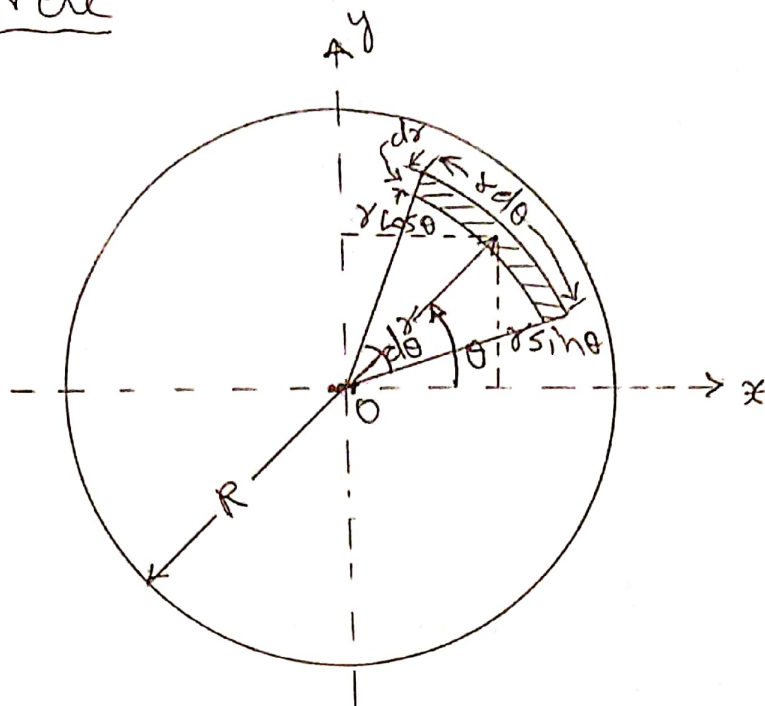
$$\text{Area of quarter circle} = \frac{\pi R^2}{4}$$

$$\therefore \bar{x} = \frac{\text{Moment of area about OY axis}}{\text{Area of quarter circle}}$$

$$= \frac{R^3}{3} / \frac{\pi R^2}{4} = \frac{R^3}{3} \times \frac{4}{\pi R^2} = \frac{4R}{3\pi}$$



iv) Circle



Consider the circle of radius 'R' and the details of elementary strip as shown in the figure.

Area of the element $= r d\theta \times dr$.

Moment of elemental area about OX-axis

$$= r d\theta \times dr \sin \theta \times r$$

$$= r^2 dr \sin \theta d\theta$$

Moment of whole area about the ~~OX-axis~~

$$\text{OX-axis, } \bar{y} = \int_0^R r^2 dr \times \int_0^{2\pi} \sin \theta d\theta$$

$$= \left[\frac{r^3}{3} \right]_0^R \left[-\cos \theta \right]_0^{2\pi}$$

$$= \frac{R^3}{3} \times [-1 - (-1)]$$

$$= 0$$

$$\therefore \bar{y} = 0$$

Moment of area about y-axis

$$\bar{x} = \int_0^R r^2 dr \times \int_0^{2\pi} \sin \theta d\theta$$

$$= \left[\frac{r^3}{3} \right]_0^R \times \left[\sin \theta \right]_0^{2\pi}$$

$$= \frac{R^3}{3} \times (0 - 0)$$

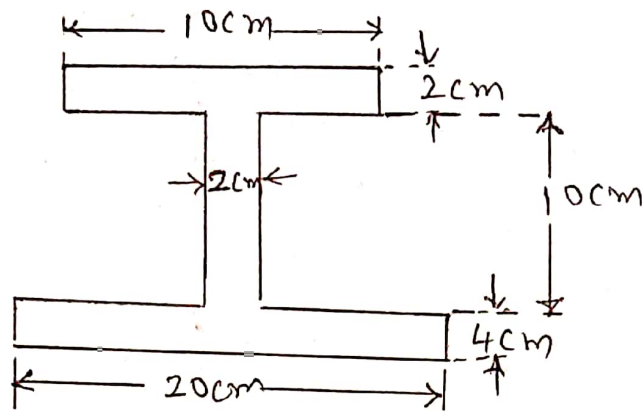
$$= 0$$

$$\therefore \bar{x} = 0$$

Hence centroid lies at the center of circle.
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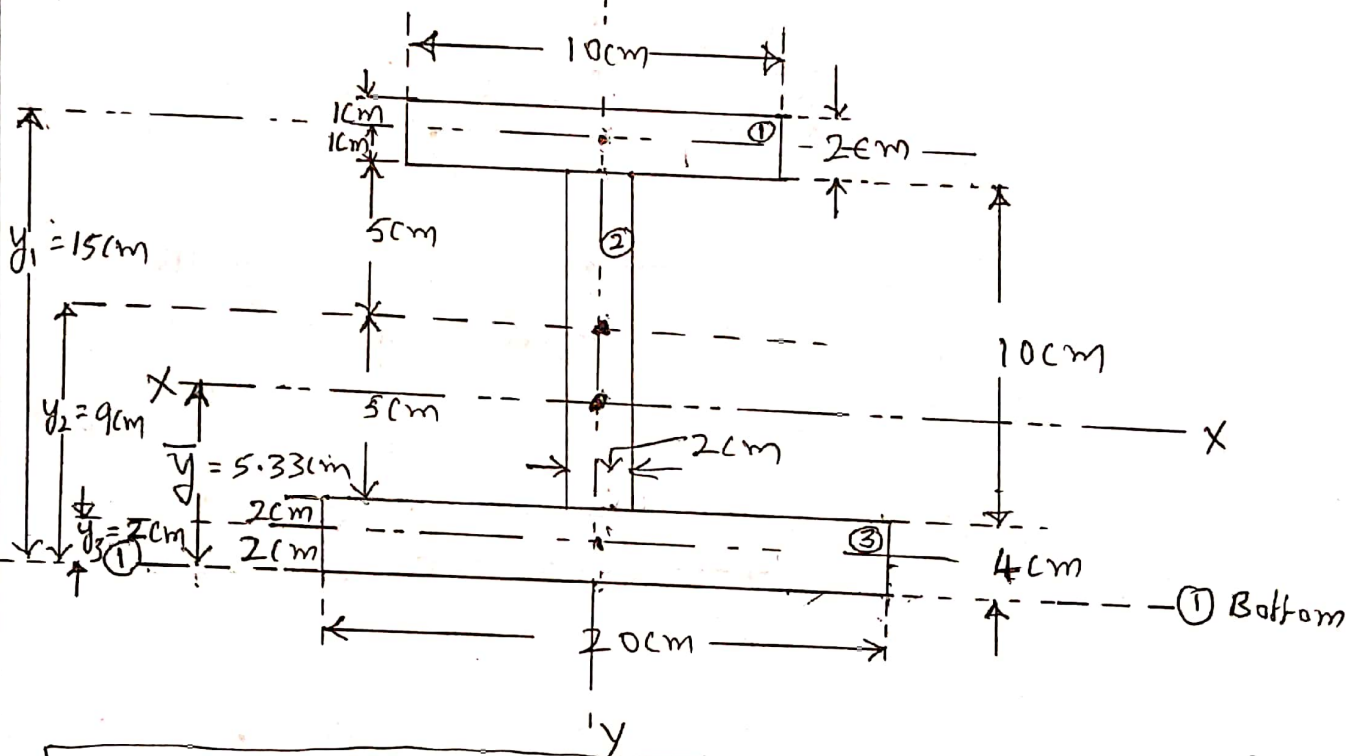
Problem

1) Determine centroid of given area shown in fig below.



Solution:-

X ← Axis of symmetry

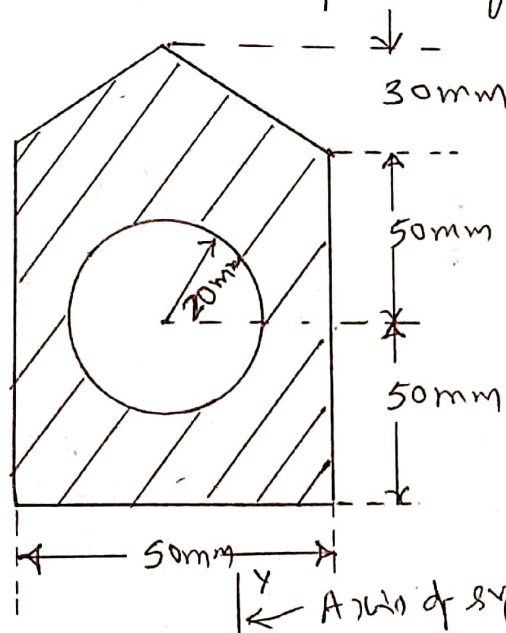


$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3}{a_1 + a_2 + a_3}$$

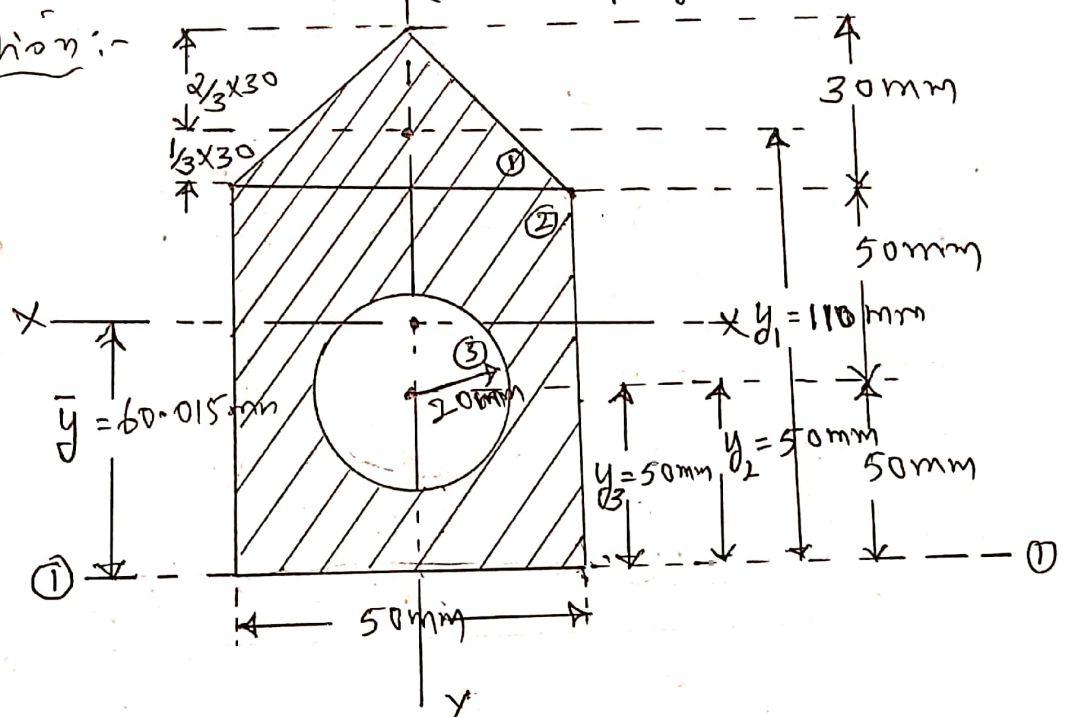
$$\begin{aligned} a_1 &= 10 \times 2 = 20 \text{ cm}^2 \\ a_2 &= 10 \times 2 = 20 \text{ cm}^2 \\ a_3 &= 20 \times 4 = 80 \text{ cm}^2 \end{aligned}$$

$$\bar{y} = \frac{20 \times 15 + 20 \times 9 + 80 \times 2}{20 + 20 + 80} = \underline{\underline{5.33 \text{ cm}}}$$

2) Determine centroid of the given area. (hatched portion)



Solution:-



We have

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 - a_3 y_3}{a_1 + a_2 - a_3}$$

$$a_1 = \frac{1}{2} \times 50 \times 30 = 750 \text{ mm}^2$$

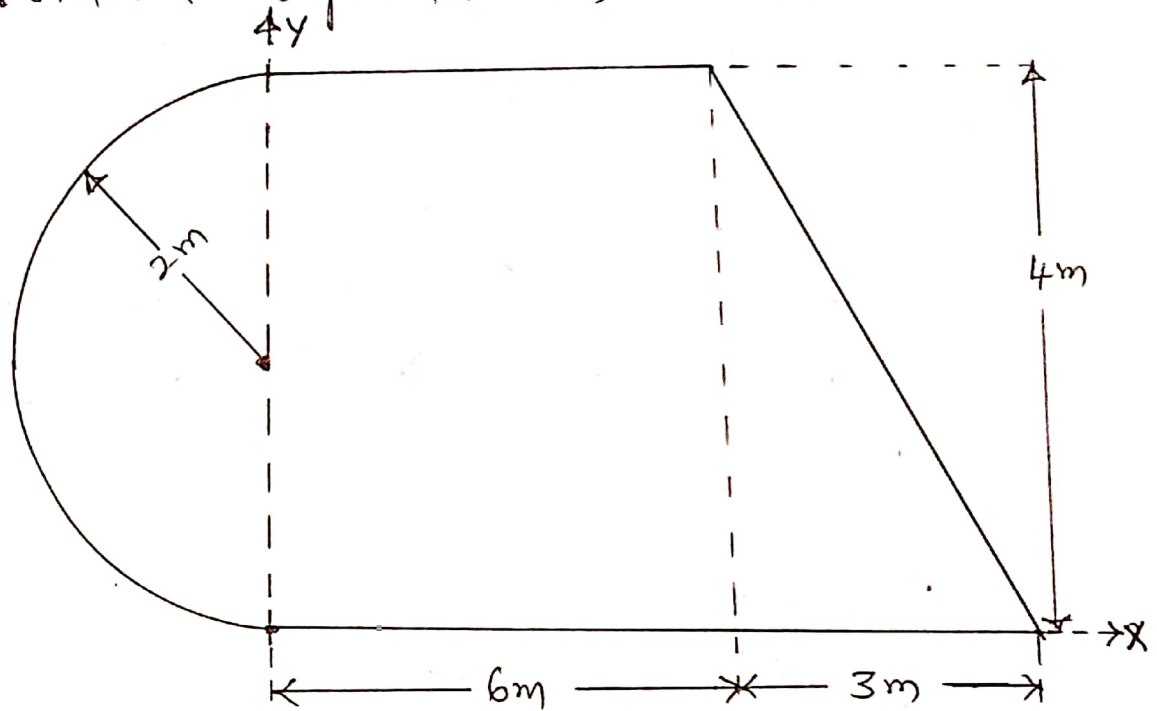
$$a_2 = 50 \times 100 = 5000 \text{ mm}^2$$

$$a_3 = \pi (20)^2 = 1256.64 \text{ mm}^2$$

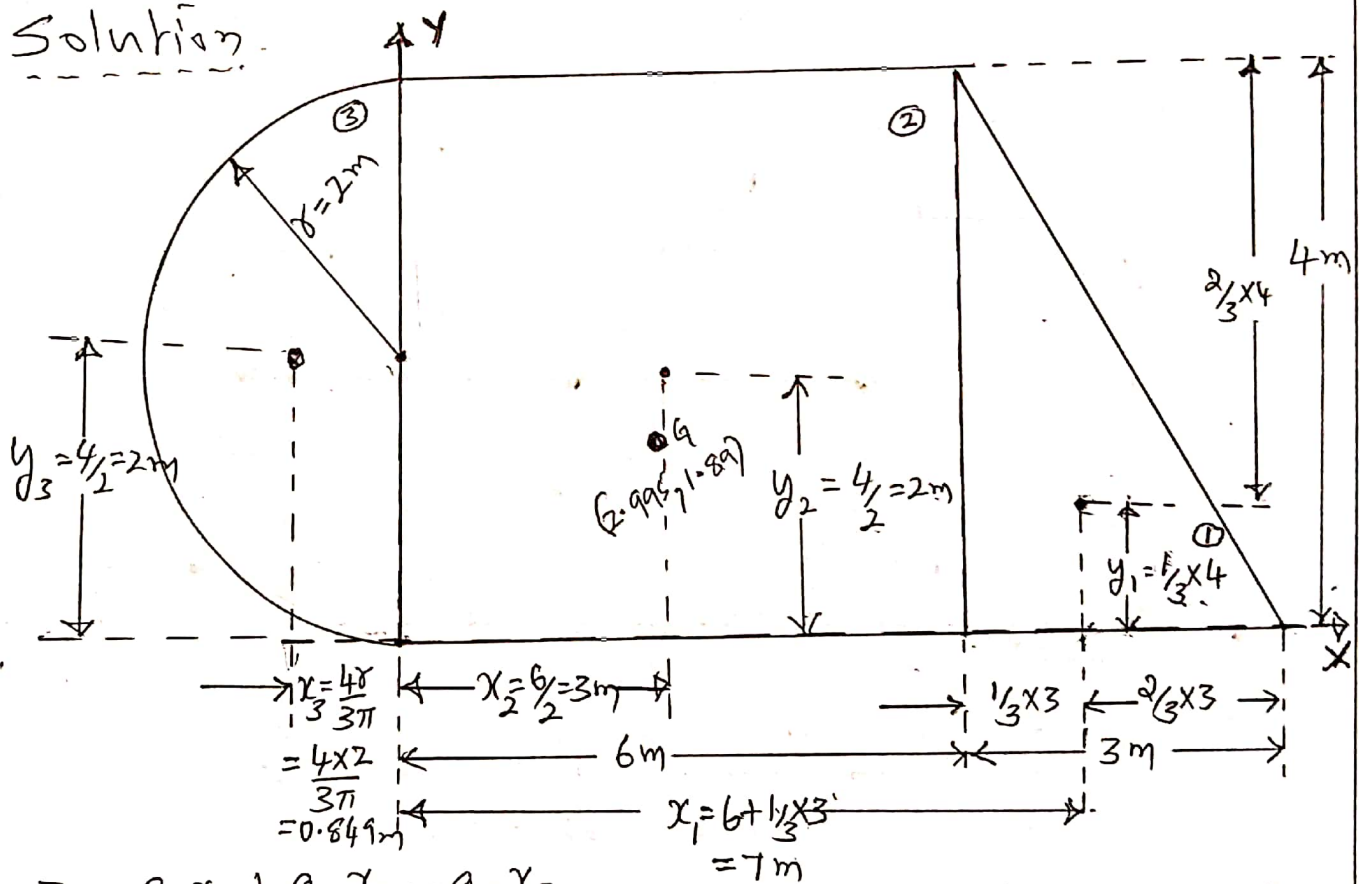
$$\bar{y} = \frac{750 \times 110 + 5000 \times 50 - 1256.64 \times 50}{750 + 5000 - 1256.64}$$

$$= \underline{\underline{60.015 \text{ mm}}}$$

③ Determine the centroid of the area shown in fig. from the reference axis x and y .



Solution.



$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 - a_3 x_3}{a_1 + a_2 - a_3}$$

$$= \frac{6 \times 7 + 24 \times 3 - (2 \times 3 \cdot 14) \times 0.849}{6 + 24 - (2 \times 3 \cdot 14)} = 2.995 \text{ m}$$

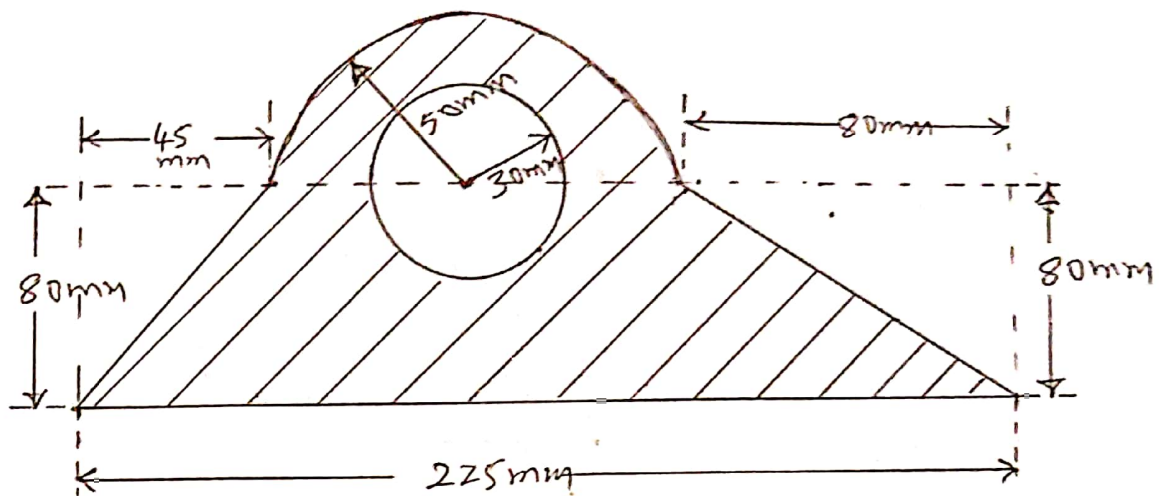
$$a_1 = \frac{1}{2} \times 3 \times 4 = 6 \text{ m}^2$$

$$a_2 = 6 \times 4 = 24 \text{ m}^2$$

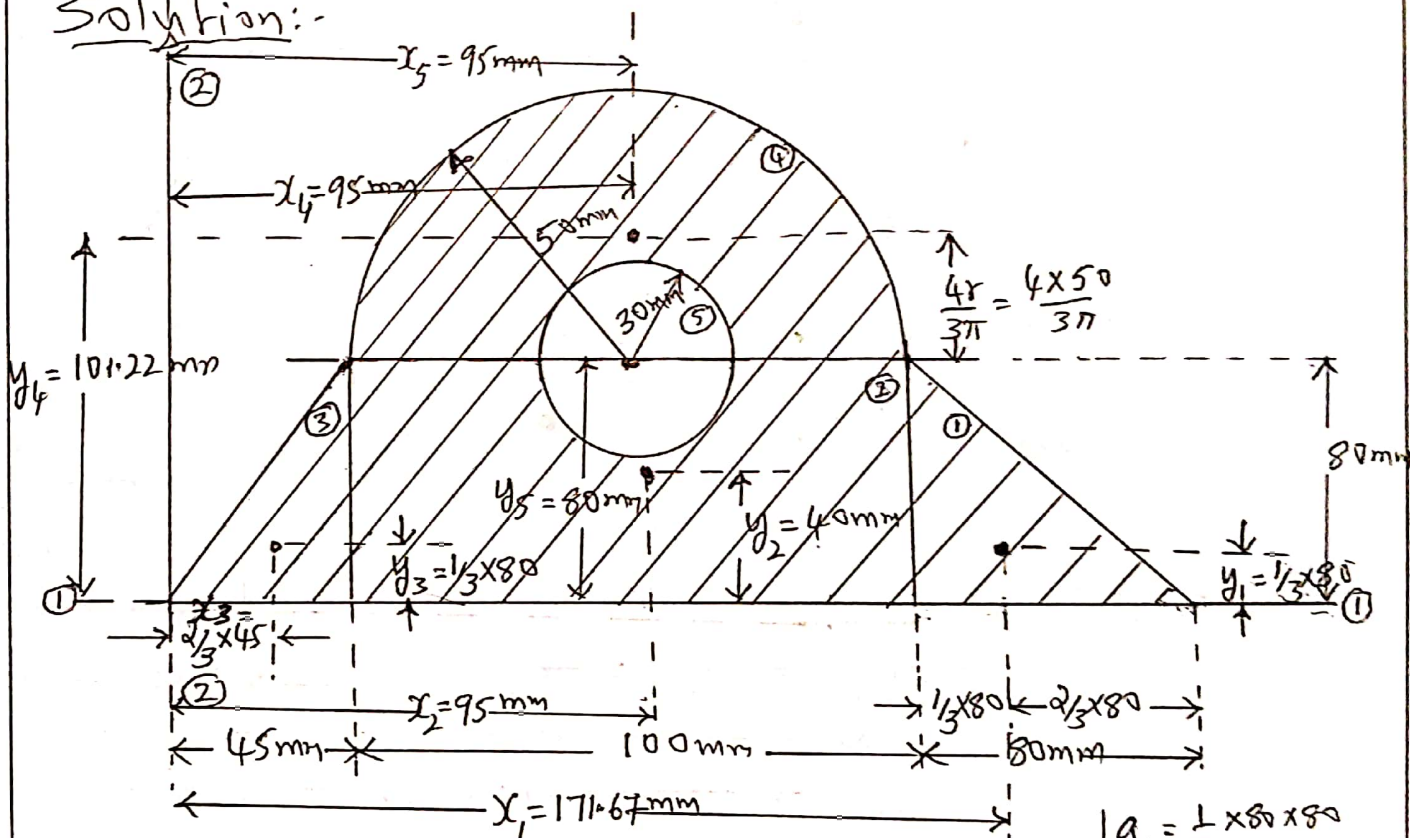
$$a_3 = \frac{\pi r^2}{2} = \frac{\pi \cdot 2^2}{2} = 2\pi \text{ m}^2 = 2 \times 3 \cdot 14 \text{ m}^2$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3}{a_1 + a_2 + a_3} = \frac{6 \times \frac{1}{3} \times 4 + 24 \times 2 + (2 \times 3 \cdot 14) \times 2}{6 + 24 + (2 \times 3 \cdot 14)} = 1.89 \text{ m}$$

4) Locate the centroid of the plane area shown in fig (Hatched portion only)



Solution:-



$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 + a_3 x_3 + a_4 x_4 - a_5 x_5}{a_1 + a_2 + a_3 + a_4 - a_5}$$

$$= \frac{3200 \times 171.67 + 8000 \times 95 + 1800 \times \frac{2}{3} \times 45 + 3926.99 \times 95 - 2827.43(95)}{3200 + 8000 + 1800 + 3926.99 - 2827.43}$$

$$= 109.103 \text{ mm}$$

$$a_1 = \frac{1}{2} \times 80 \times 80 = 3200 \text{ mm}^2$$

$$a_2 = 100 \times 80 = 8000 \text{ mm}^2$$

$$a_3 = \frac{1}{2} \times 45 \times 80 = 1800 \text{ mm}^2$$

$$a_4 = \frac{\pi r^2}{2} = \frac{\pi (50)^2}{2} = 3926.99 \text{ mm}^2$$

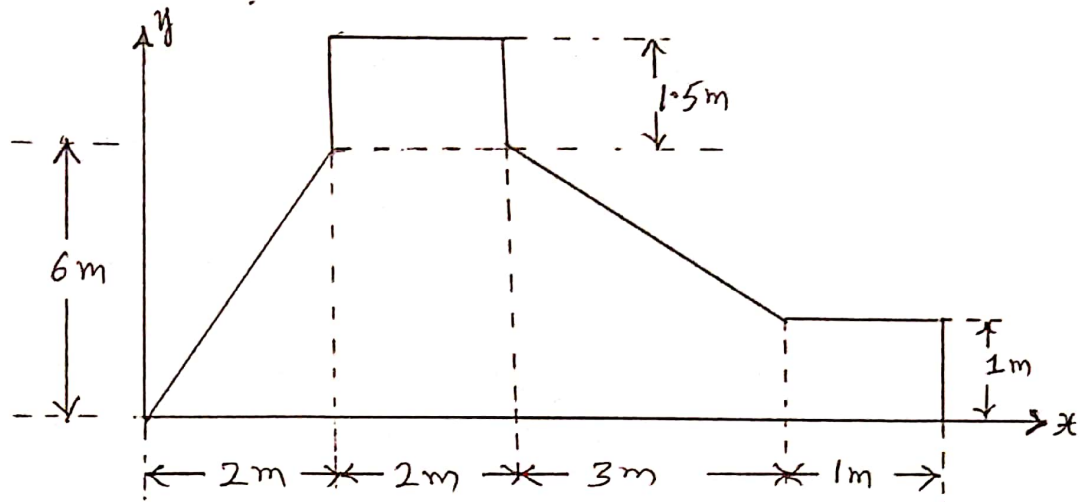
$$a_5 = \pi r^2 = \pi (30)^2 = 2827.43 \text{ mm}^2$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3 + a_4 y_4 - a_5 y_5}{a_1 + a_2 + a_3 + a_4 - a_5}$$

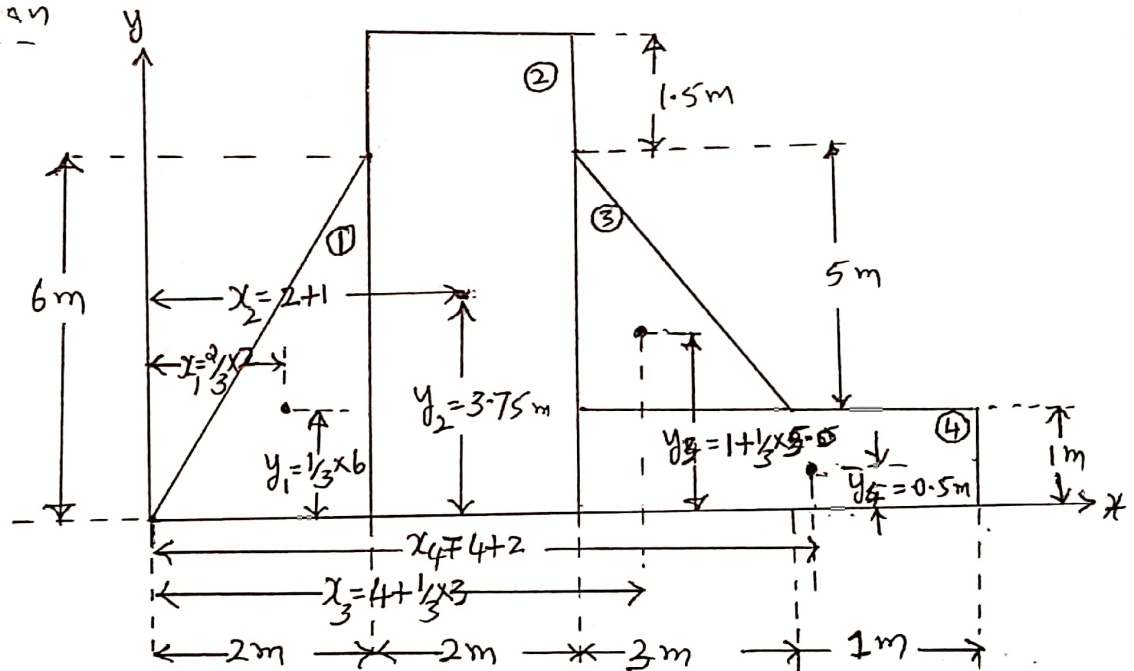
$$= \frac{3200 \times \frac{1}{3} \times 80 + 8000 \times 40 + 1800 \times \frac{1}{2} \times 80 + 3926.99 \times 101.22 - 2827.43 \times 80}{3200 + 8000 + 1800 + 3926.99 - 2827.43}$$

$$= \underline{\underline{44.3 \text{ mm}}}$$

5) Locate the centroid of the section of concrete dam shown in fig



Solution



$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 + a_3 x_3 + a_4 x_4}{a_1 + a_2 + a_3 + a_4}$$

$$= \frac{6 \times \left(\frac{2}{3} \times 2\right) + 15(3) + 7.5\left(4 + \frac{1}{3} \times 3\right) + 4(4+2)}{6 + 15 + 7.5 + 4}$$

$$= \underline{\underline{3.523m}}$$

$$a_1 = \frac{1}{2} \times 2 \times 6 = 6m^2$$

$$a_2 = 7.5 \times 2 = 15m^2$$

$$a_3 = \frac{1}{2} \times 3 \times 5 = 7.5m^2$$

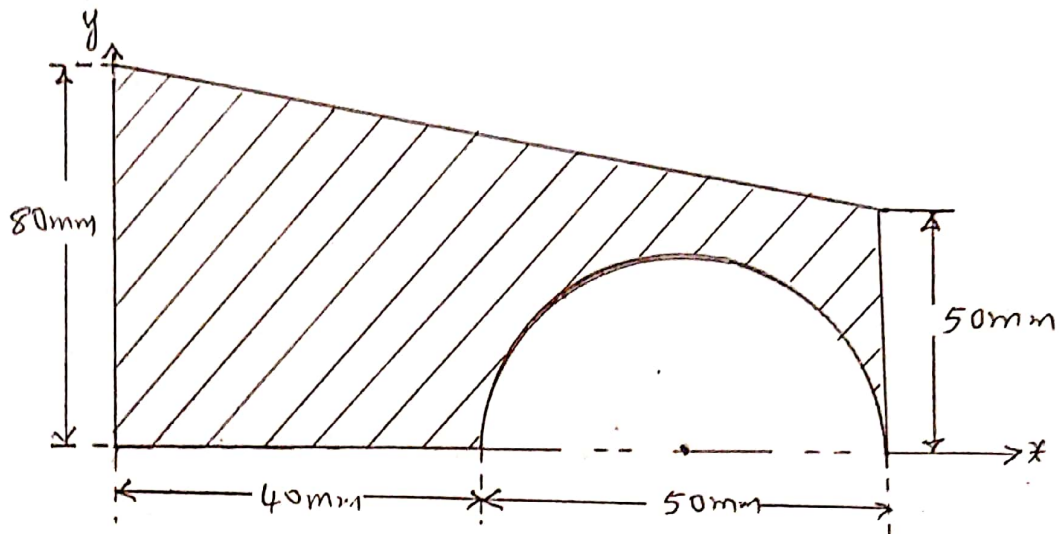
$$a_4 = 4 \times 1 = 4m^2$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3 + a_4 y_4}{a_1 + a_2 + a_3 + a_4}$$

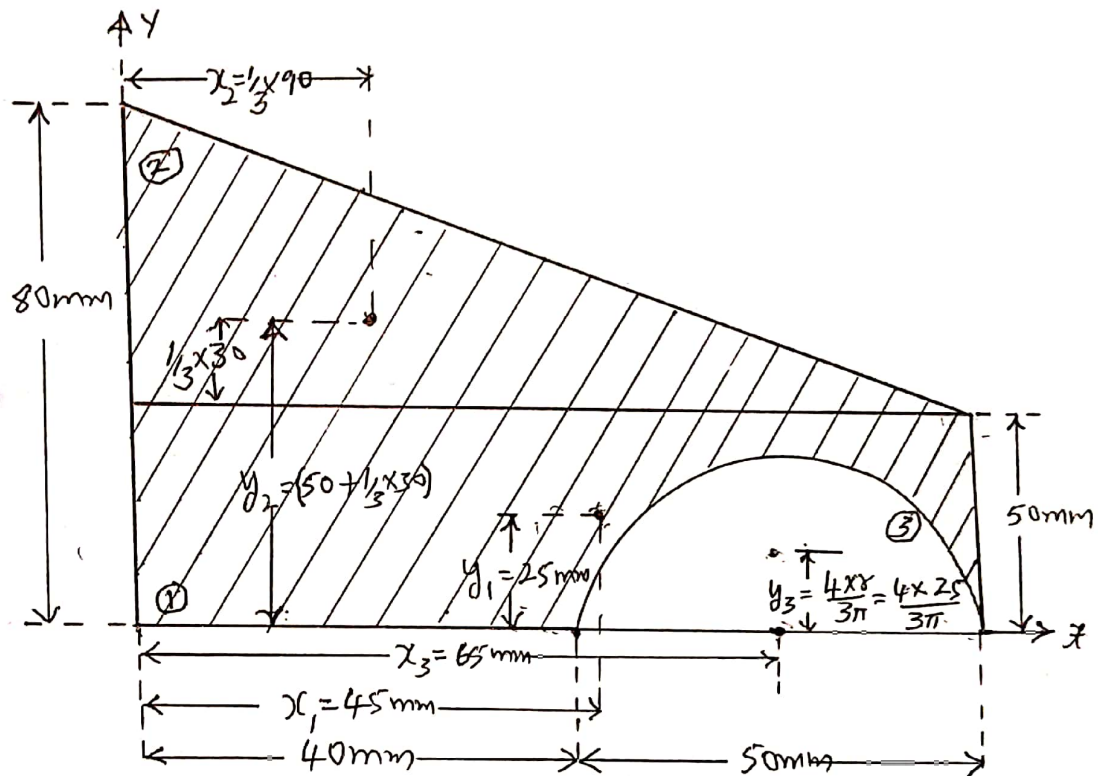
$$= \frac{6 \times \left(\frac{1}{3} \times 6\right) + 15(3.75) + 7.5\left(1 + \frac{1}{3} \times 5\right) + 4(0.5)}{6 + 15 + 7.5 + 4}$$

$$= \underline{\underline{2.776m}}$$

5) A semicircular area is removed from a trapezium as shown in figure. Determine the centroid of the remainder.



Solution:



$$a_1 = 90 \times 50 = 4500 \text{ mm}^2 \quad a_2 = \frac{1}{2} \times 90 \times 30 = 1350 \text{ mm}^2$$

$$a_3 = \frac{\pi r^2}{2} = \frac{\pi (25)^2}{2} = 981.75 \text{ mm}^2$$

$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 - a_3 x_3}{a_1 + a_2 - a_3} = \frac{4500 \times 45 + 1350 \times \frac{1}{3} \times 90 - 981.75 \times 65}{4500 + 1350 - 981.75}$$

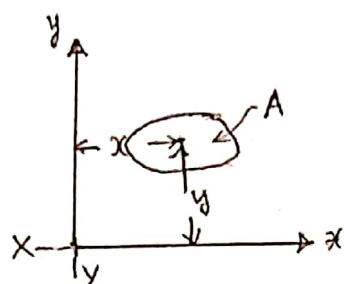
$$= \underline{\underline{47.77 \text{ mm}}}$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 - a_3 y_3}{a_1 + a_2 - a_3} = \frac{4500 \times 25 + 1350 \times (50 + \frac{1}{3} \times 30) - 981.75 \times (\frac{4 \times 25}{3\pi})}{4500 + 1350 - 981.75}$$

$$= \underline{\underline{37.61 \text{ mm}}}$$

Moment of Inertia

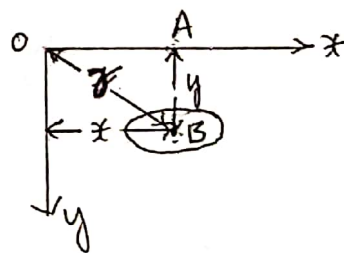
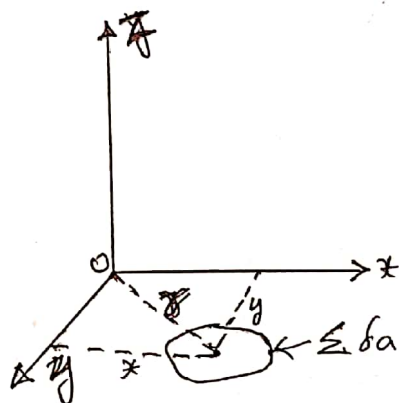
The second moment of area about any reference axis is known as moment of inertia about that axis.



The second moment of area about yy -axis $= A x^2 = I_{yy}$

The second moment of area about xx -axis $= A y^2 = I_{xx}$

Perpendicular axis theorem:-



It states that if I_{xx} and I_{yy} are the moment of inertia of an area about two mutually perpendicular axes yy and xx in plane, the moment of inertia about axis zz perpendicular to plane is given by, $I_{zz} = I_{xx} + I_{yy}$.

Proof:

Consider an area da in x - y plane.

Let x = perpendicular distance of C.G. of area from yy axis

y = Perpendicular distance of C.G. of area from OY axis

x = Perpendicular distance of C.G. of area from OZ -axis

Moment of inertia of area about $Ox = I_{xx} = \sum da x^2$ - (1)

Moment of inertia of area about $Oy = I_{yy} = \sum da y^2$ - (2)

Moment of inertia about OZ axis = $I_{zz} = \sum da r^2$ - (3)

But From ΔOAB .

$$r^2 = x^2 + y^2 \quad - (4)$$

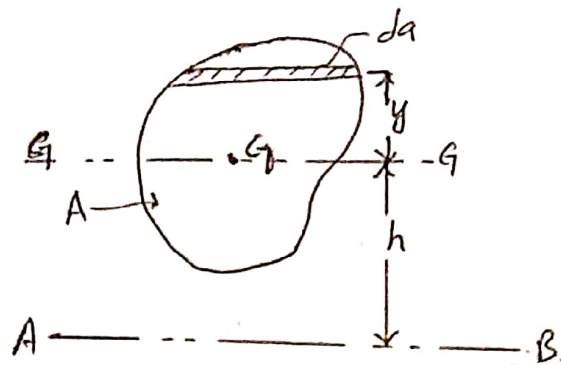
Substitute (4) in (3)

$$I_{zz} = \sum da (x^2 + y^2)$$

$$I_{zz} = \sum da x^2 + \sum da y^2 \quad - (5)$$

$$\therefore \boxed{I_{zz} = I_{xx} + I_{yy}}$$

Parallel axis theorem:-



It states that "If I_{GG} is the moment of inertia of an area about CG axis, the moment of inertia of the same area about an axis AB parallel to centroidal axis GG is given by

$$I_{AB} = I_{GG} + Ah^2.$$

Where A = Area of the element

h = perpendicular distance between AB and GG.

Proof: Consider an area 'A' whose centroid is at 'G'. Let AB is the axis parallel to centroidal axis GG, at a perpendicular distance 'h' from GG.

M.I of elementary strip about AB = $da(h+y)^2$

$$= day^2 + dah^2 + 2xhxy$$

M.I of whole area about AB = $\sum day^2 + \sum dah^2 + \sum 2xhxy$

In above eqn, $\sum day^2 = I_{GG}$, $\sum dah^2 = Ah^2$.

and $\sum 2xhxy$, $y=0$,

$$\therefore \sum 2xhxy = 0$$

$$\therefore I_{AB} = I_{GG} + Ah^2.$$

Polar moment of inertia:-

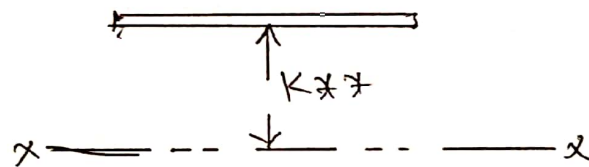
Moment of inertia of an area about an axis perpendicular to the plane is called polar moment of inertia. It is denoted by I_{zz} .

$$\therefore I_{zz} = I_{xx} + I_{yy}.$$

Radius of gyration:-

It can be defined as distance from any reference axis where the whole area is assumed to be squeezed (concentrated).

Mathematically $K = \sqrt{I/A}$



$$\therefore \text{Radius of gyration about } x\text{-axis} = K_{xx} = \sqrt{\frac{I_{xx}}{A}}$$

Radius of gyration about y -axis

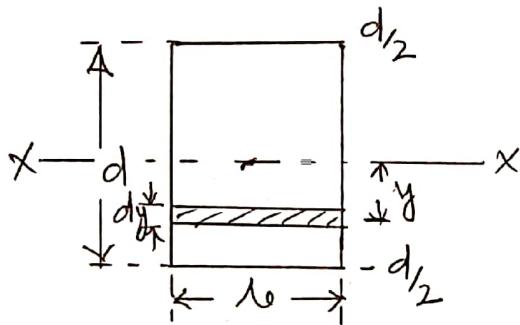
$$K_{yy} = \sqrt{\frac{I_{yy}}{A}}$$

Moment of inertia of regular figures:-

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i) Rectangle:-

Consider a rectangle of width 'b' and depth 'd'. Let us consider an elementary area of thickness 'dy' at a distance of 'y' from centroidal ~~xxxxx~~ XX-axis.



Area of elementary strip = $b \times dy$

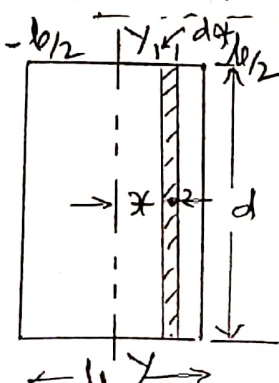
Second moment of elementary strip about XX-axis = $b \times dy \times y^2$

Moment of inertia of whole area about XX-axis can be determined by integrating above eqn. between limits $d/2$ to $-d/2$.

$$I_{xx} = \int_{-d/2}^{d/2} b \times dy \times y^2 = b \left[\frac{y^3}{3} \right]_{-d/2}^{d/2} = \frac{b}{3} \left[\frac{d^3}{8} + \frac{d^3}{8} \right] = \frac{bd^3}{12}$$

$$\therefore \boxed{I_{xx} = \frac{bd^3}{12}}$$

Moment of inertia about YY-axis



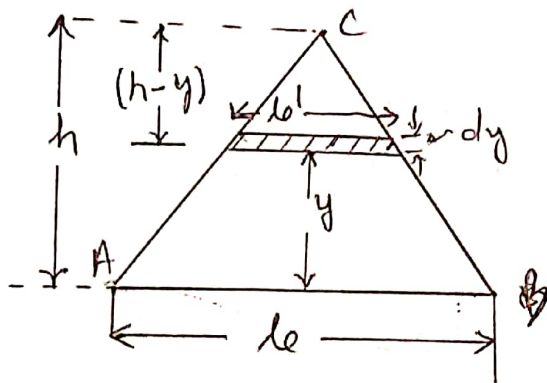
Moment of inertia of elementary area about YY-axis = $d \times dx \times x^2$

$\therefore$ Moment of inertia about YY-axis = $\int_{-b/2}^{b/2} d \cdot dx \cdot x^2$

$$I_{yy} = \int_{-b/2}^{b/2} d \cdot dx \cdot x^2 = d \left[\frac{x^3}{3} \right]_{-b/2}^{b/2} = \frac{d}{3} \left[\frac{b^3}{8} + \frac{b^3}{8} \right] = \frac{db^3}{12}$$

$$\therefore \boxed{I_{yy} = \frac{db^3}{12}}$$

ii) Triangle:-



Consider a triangle of base width 'b' and height 'h'. Consider an elementary area parallel to base and at a distance of 'y' from base. Let the thickness of elementary strip is 'dy'.

From $\Delta^e ABC$

$$\frac{b'}{b} = \frac{(h-y)}{h}$$

$$b' = \frac{b}{h} (h-y) = b \left(1 - \frac{y}{h}\right)$$

Area of elementary strip = $b \left(1 - \frac{y}{h}\right) \times dy$

Moment of elementary strip about base
 $= b \left(1 - \frac{y}{h}\right) \times y^2 dy = b \left(y^2 - \frac{y^3}{h}\right) dy$

M.I of whole triangle about base

$$I_{AB} = \int_0^h b \left(y^2 dy - \frac{y^3}{h} dy\right)$$

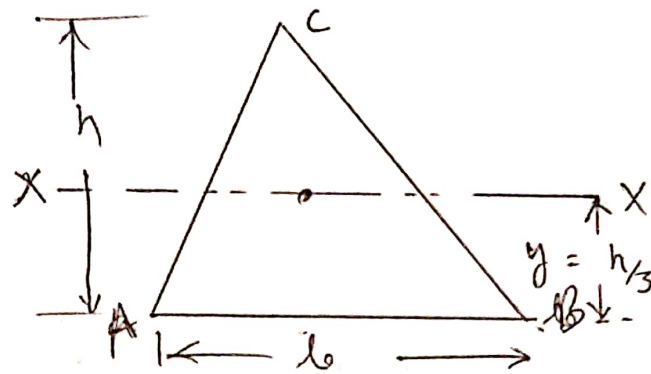
$$= b \left[\frac{y^3}{3} - \frac{y^4}{4h} \right]_0^h = b \left[\frac{h^3}{3} - \frac{h^4}{4h} \right]$$

$$\boxed{I_{AB} = \frac{bh^3}{12}}$$

Moment of inertia.

M.I of triangle about centroidal xx-axis

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From parallel axis theorem,

$$I_{AB} = I_{xx} + Ay^2$$

$$\therefore I_{xx} = I_{AB} - Ay^2$$

$$= \frac{bh^3}{12} - \frac{1}{2} \times b \times h \left(\frac{h}{3}\right)^2$$

$$= \frac{bh^3}{12} - \frac{1}{2} \times b \times h \times \frac{h^2}{9}$$

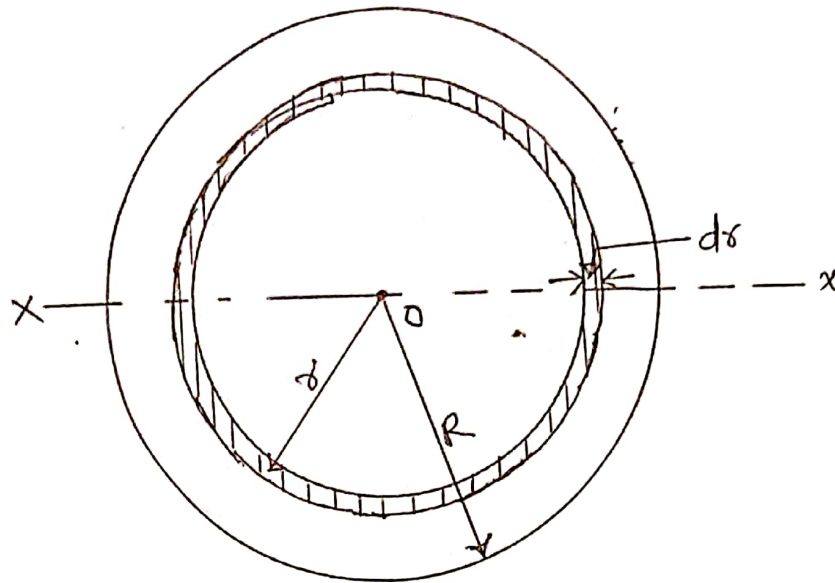
$$= \frac{bh^3}{12} - \frac{bh^3}{18}$$

$$= \frac{3bh^3 - 2bh^3}{36}$$

$$= \frac{bh^3}{36}$$

$$\therefore I_{xx} = \frac{bh^3}{36}$$

iii) Circle



Consider a circle of radius 'R'. 'r' be the radius of elementary strip of thickness 'dr'.

Area of elementary strip = $2\pi r \times dr$.

M.I of elementary strip about OZ axis.

$$= \text{Area} \times \text{distance}^2 = 2\pi r \times dr \times r^2 = 2\pi r^3 dr.$$

Moment of inertia of whole area about OZ axis

$$I_{ZZ} = \int_0^R 2\pi r^3 dr$$

$$\therefore I_{ZZ} = 2\pi \left(\frac{r^4}{4} \right)_0^R = 2\pi \frac{R^4}{4} = \frac{\pi R^4}{2}$$

But we know that, ~~R = D~~ $R = \frac{D}{2}$

$$\therefore I_{ZZ} = \frac{\pi \left(\frac{D}{2} \right)^4}{2} = \frac{\pi D^4}{32}$$

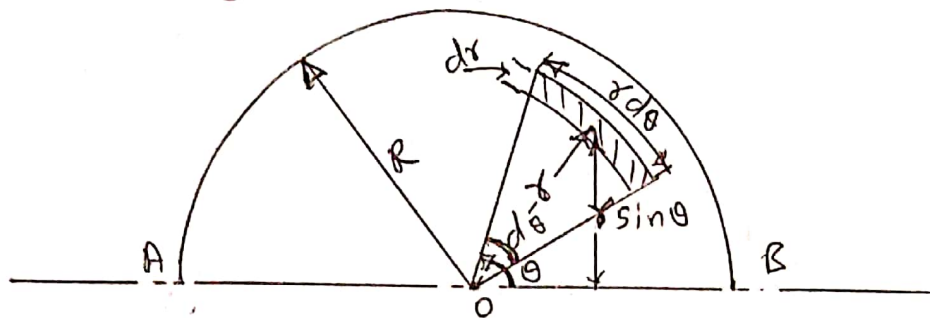
Using perpendicular axis theorem,

$$I_{ZZ} = I_{xx} + I_{yy}, \text{ Here } I_{xx} = I_{yy}$$

$$I_{ZZ} = 2I_{xx}$$

$$I_{xx} = \frac{I_{ZZ}}{2} = \frac{\pi D^4}{32 \times 2} = \frac{\pi D^4}{64} = I_{yy}$$

iv) Semicircle



Consider a semicircle of radius R .

Let r be the radius of elementary strip of thickness dr from O .

Area of elementary strip $= r d\theta \times dr$.

$$\begin{aligned} \text{M.I. of elementary strip} &= \text{Area} \times \text{distance}^2 \\ &= r d\theta \cdot dr \times r^2 \sin^2 \theta \end{aligned}$$

$\therefore$ Moment of inertia of whole area about

$$\text{base AB} = \int_0^R r^3 dr \times \int_0^\pi \sin^2 \theta d\theta$$

$$= \left[\frac{r^4}{4} \right]_0^R \int_0^\pi \frac{(1 - \cos 2\theta)}{2} d\theta$$

$$= \frac{R^4}{4 \times 2} \int_0^\pi (1 - \cos 2\theta) d\theta$$

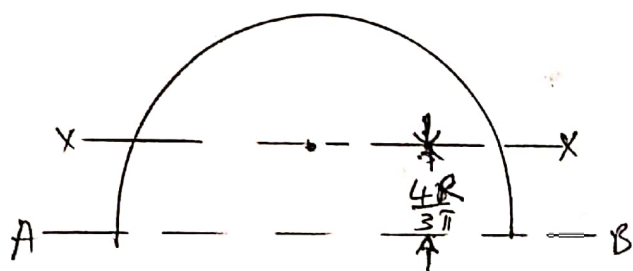
$$= \frac{R^4}{8} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^\pi$$

$$= \frac{R^4}{8} [\pi - 0 + 0 - 0]$$

$$= \frac{\pi R^4}{8}$$

$$(R = \frac{D}{2})$$

$$I_{AB} = \frac{\pi}{8} \left(\frac{D}{2} \right)^4 = \frac{\pi D^4}{128}$$

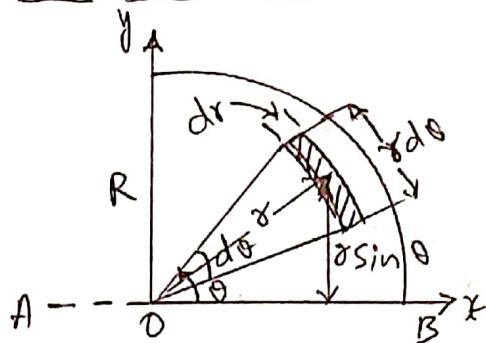


Using parallel axis theorem

$$\begin{aligned} I_{XX} &= \frac{\pi R^4}{8} - \frac{\pi R^2}{2} \left(\frac{4R}{3\pi} \right)^2 \\ &= 0.39269 R^4 - 0.2829 R^4 \end{aligned}$$

$$\therefore \underline{\underline{I_{XX} = 0.11 R^4}}$$

v) Quarter Circle:-



Consider a quarter circle of radius 'R'.

Consider an elementary strip of thickness dr , and length ' $r d\theta$ ' at a distance ' r ' from 'O'.

Area of elementary strip = $r d\theta \times dr$.

Moment of elementary strip about base AB = Area * distance²

$$= r d\theta \times dr \times r^2 \sin^2 \theta = r^3 dr \sin^2 \theta d\theta$$

Moment of inertia of whole area about base

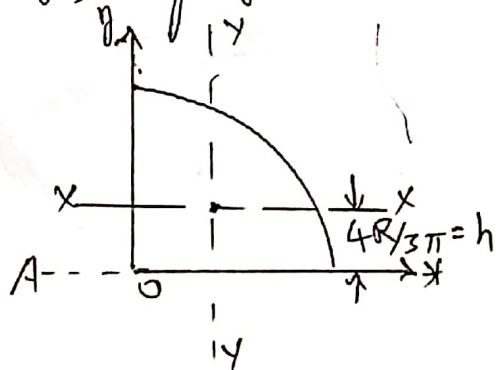
$$I_{AB} = \int_0^R r^3 dr \int_0^{\pi/2} \sin^2 \theta d\theta$$

$$= \left[\frac{r^4}{4} \right]_0^R \times \int_0^{\pi/2} (1 - \frac{\cos 2\theta}{2}) d\theta$$

$$= \frac{R^4}{8} \left(\theta - \frac{\sin 2\theta}{2} \right)_0^{\pi/2}$$

$$= \frac{R^4}{8} \left(\frac{\pi}{2} - 0 + 0 - 0 \right) = \frac{\pi R^4}{16}$$

Using parallel axis theorem



$$I_{AB} = I_{xx} + A x^2$$

$$\therefore I_{xx} = I_{AB} - A x^2$$

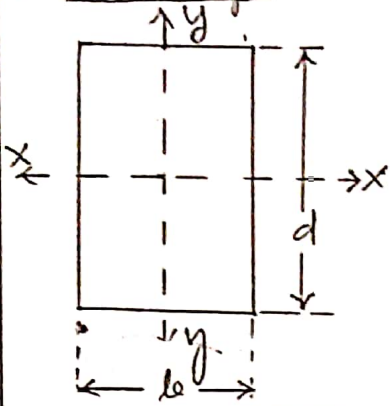
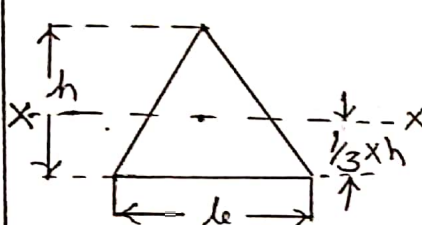
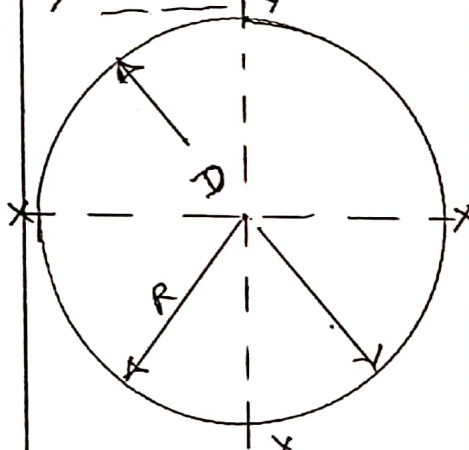
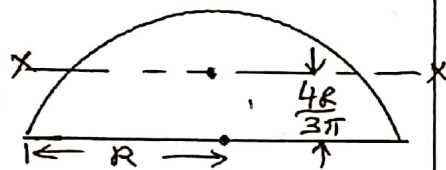
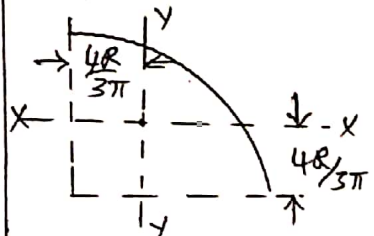
$$= \frac{\pi R^4}{16} = \frac{\pi R^2}{4} \times \left(\frac{4R}{3\pi} \right)^2$$

$$= 0.196 R^4 - 0.141 R^4$$

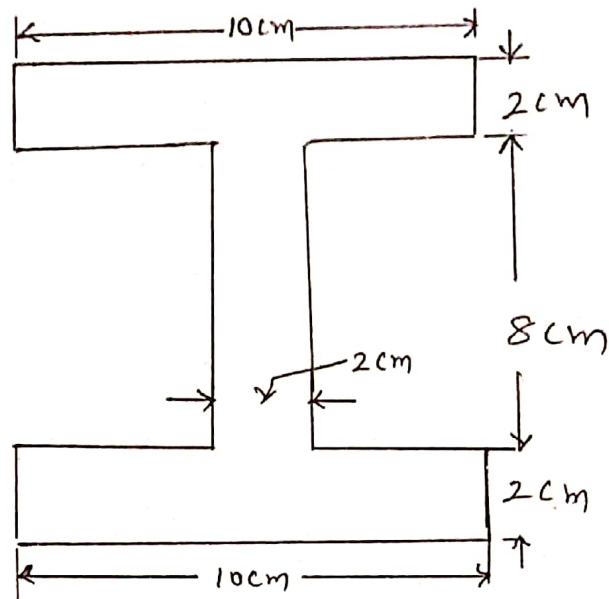
$$= 0.055 R^4$$

$$\therefore \underline{I_{xx} = I_{yy} = 0.055 R^4}$$

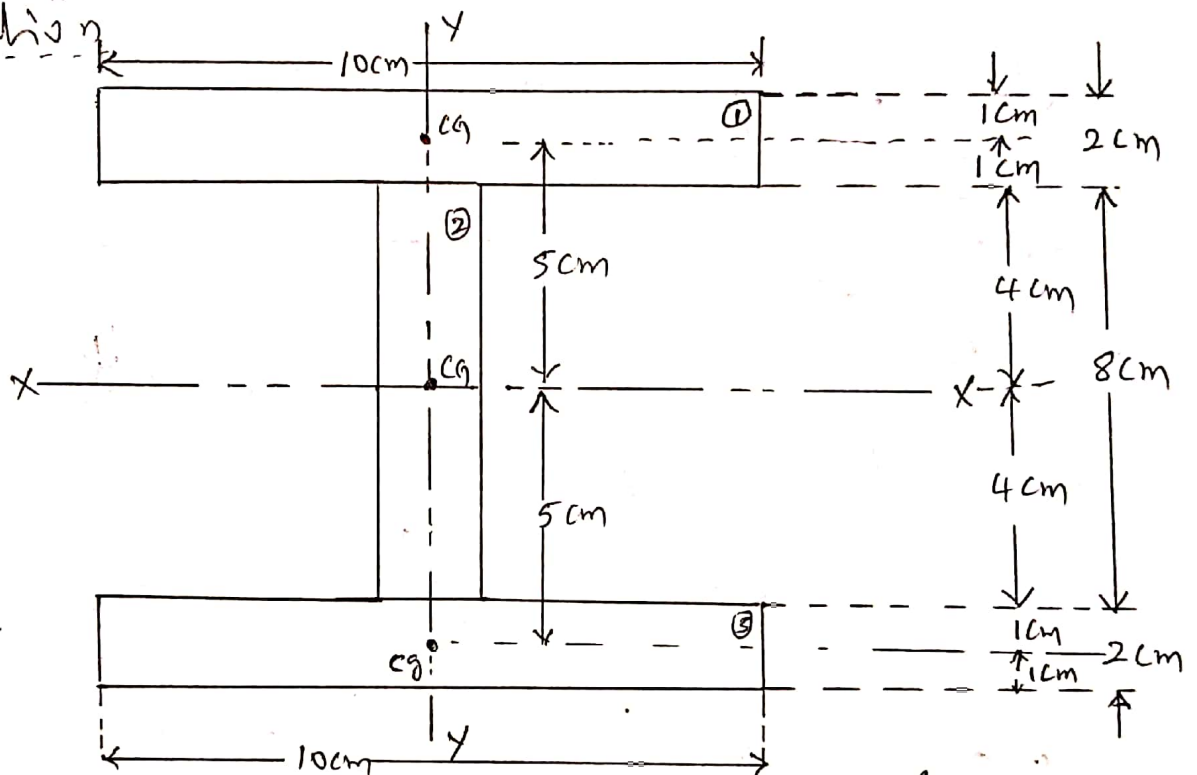
Moment of Inertia of Regular shapes:

Shape	Moment of Inertia (M.I)
1) <u>Rectangle</u> 	$I_{xx} = \frac{ld^3}{12}$ $I_{yy} = \frac{dl^3}{12}$
2) <u>Triangle</u> 	$I_{xx} = \frac{bh^3}{36}$
3) <u>Circle</u> 	$I_{xx} = I_{yy} = \frac{\pi D^4}{64}$
4) <u>Semicircle</u> 	$I_{xx} = 0.11 R^4$
5) <u>Quarter circle</u> 	$I_{xx} = I_{yy} = 0.055 R^4$

1) Determine moment of inertia and radius of gyration of the section shown about centroidal XX and YY axes.



Solution



$$I_{XX} = \left[\frac{10 \times 2^3}{12} + 10 \times 2 \times (5)^2 \right] + \left[\frac{2 \times 8^3}{12} + 2 \times 8 \times (0)^2 \right] + \left[\frac{10 \times 2^3}{12} + 10 \times 2 \times (5)^2 \right] = 1098.664 \text{ cm}^4$$

$$I_{YY} = \left[\frac{2 \times 10^3}{12} + 10 \times 2 \times (0)^2 \right] + \left[\frac{8 \times 2^3}{12} + 8 \times 2 \times (0)^2 \right] + \left[\frac{10 \times 2^3}{12} + 10 \times 2 \times (0)^2 \right] = 338.66 \text{ cm}^4$$

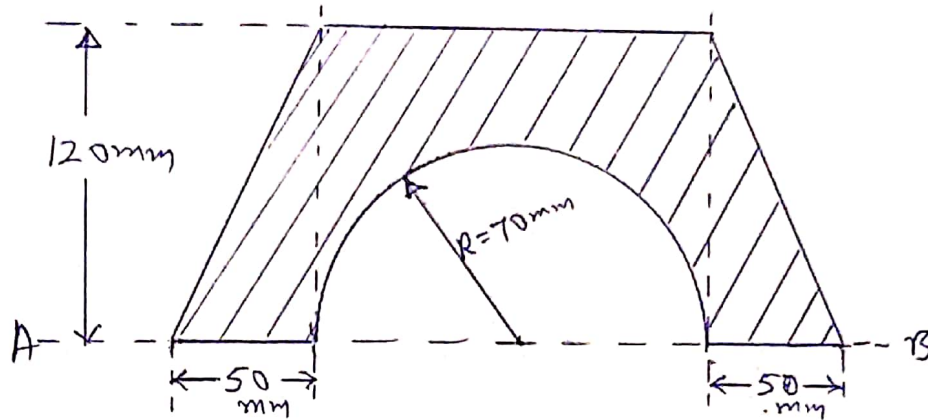
Radius of gyration $K = \sqrt{I/A}$

$$\therefore K_{xx} = \sqrt{\frac{I_{xx}}{A}} = \sqrt{\frac{1098.669}{56}} = 4.43 \text{ cm}$$

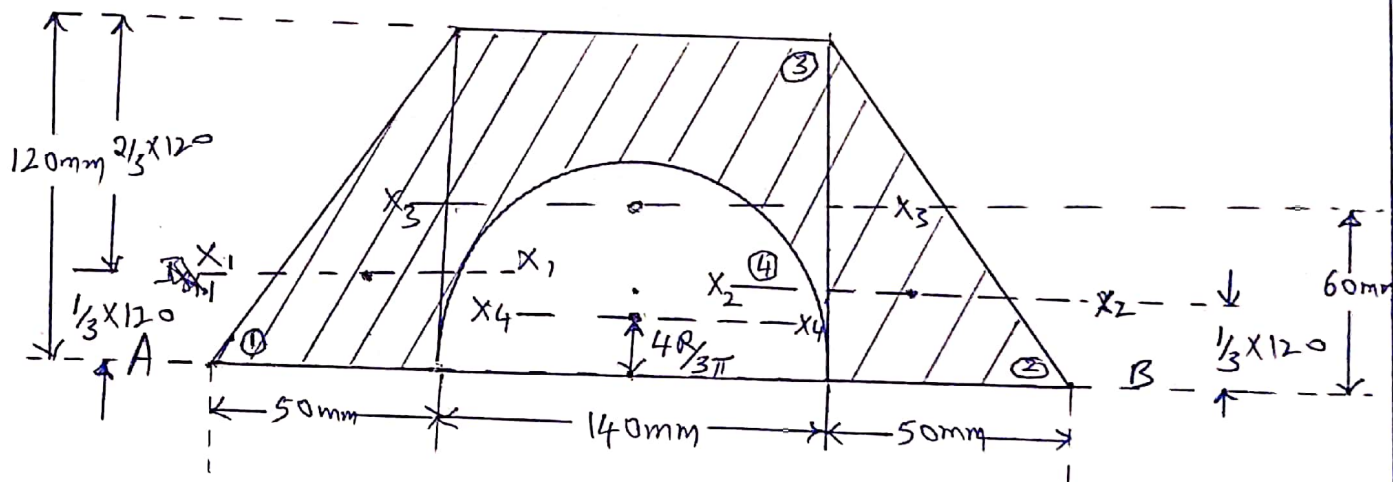
$$\begin{array}{l} \text{Area} \\ A = 10 \times 2 + 8 \times 2 + 10 \times 2 \\ = \underline{56 \text{ cm}^2} \end{array}$$

$$\therefore K_{yy} = \sqrt{\frac{I_{yy}}{A}} = \sqrt{\frac{338.66}{56}} = \underline{2.45 \text{ cm}}$$

2) Determine the moment of inertia of plane lamina about base AB. 18



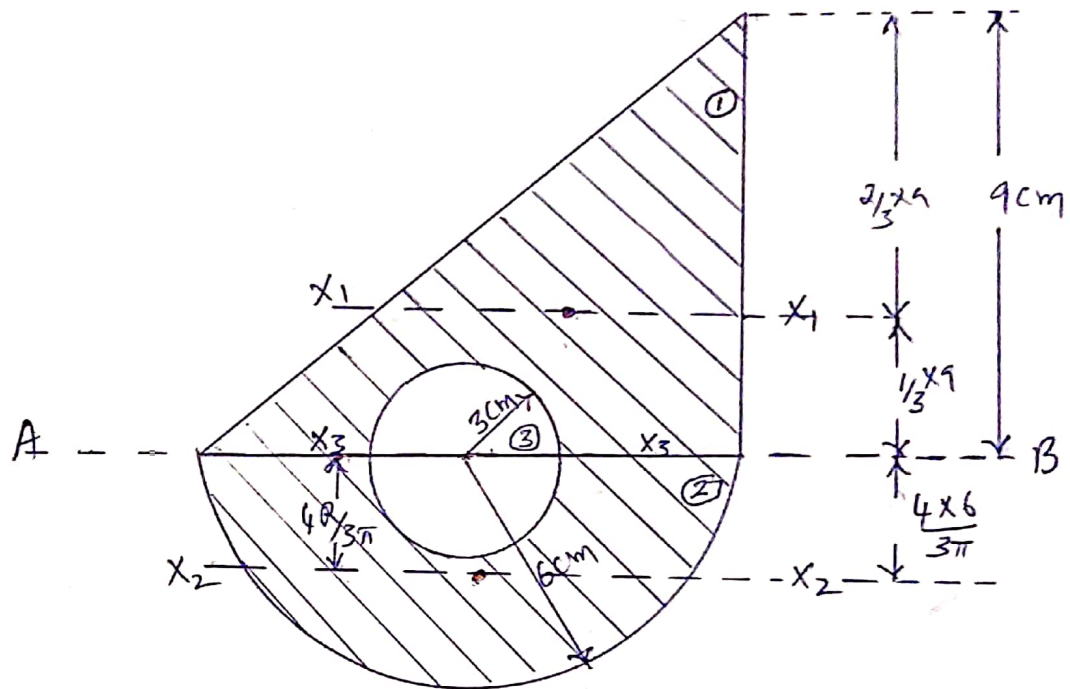
Solution:



$$\begin{aligned}
 I_{AB} = & \left[\frac{50 \times 120^3}{36} + \frac{1}{2} \times 50 \times 120 (40)^2 \right] + \left[\frac{50 \times 120^3}{36} + \frac{1}{2} \times 50 \times 120 (40)^2 \right] \\
 & + \left[\frac{140 \times 120^3}{12} + 140 \times 120 (60)^2 \right] \\
 & - \left[0.11 (70)^4 + \frac{\pi (70)^2}{2} \left(\frac{4 \times 70}{3\pi} \right)^2 \right]
 \end{aligned}$$

$$= \underline{\underline{944.50 \text{ mm}^4}}$$

3) Determine the M.I and radius of gyration for the given lamina about axis-AB. 19



$$I_{AB} = \left[\frac{12 \times 9^3}{36} + \frac{1}{2} \times 12 \times 9 \left(\frac{1}{3} \times 9 \right)^2 \right] + \left[(0.11 \times 6^4) + \frac{\pi R^2}{2} \times \left(\frac{4 \times 6}{3\pi} \right)^2 \right] - \left[\frac{\pi (6)^4}{64} + \pi \times 3^2 \times (0)^2 \right]$$

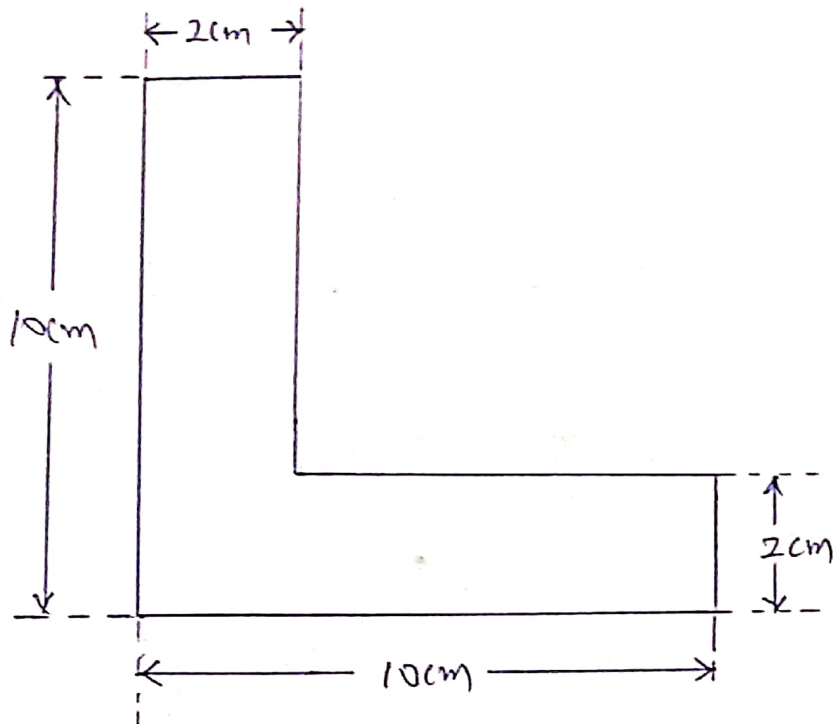
$$= \underline{\underline{1174.32 \text{ cm}^2}}$$

Radius of gyration $= K_{AB} = \sqrt{\frac{I_{AB}}{A}}$

$$A = \frac{\pi (6)^2}{2} + \frac{1}{2} \times 12 \times 9 - \pi (3)^2 =$$

Radius of gyration $K_{AB} = \sqrt{\frac{1174.32}{A}}$

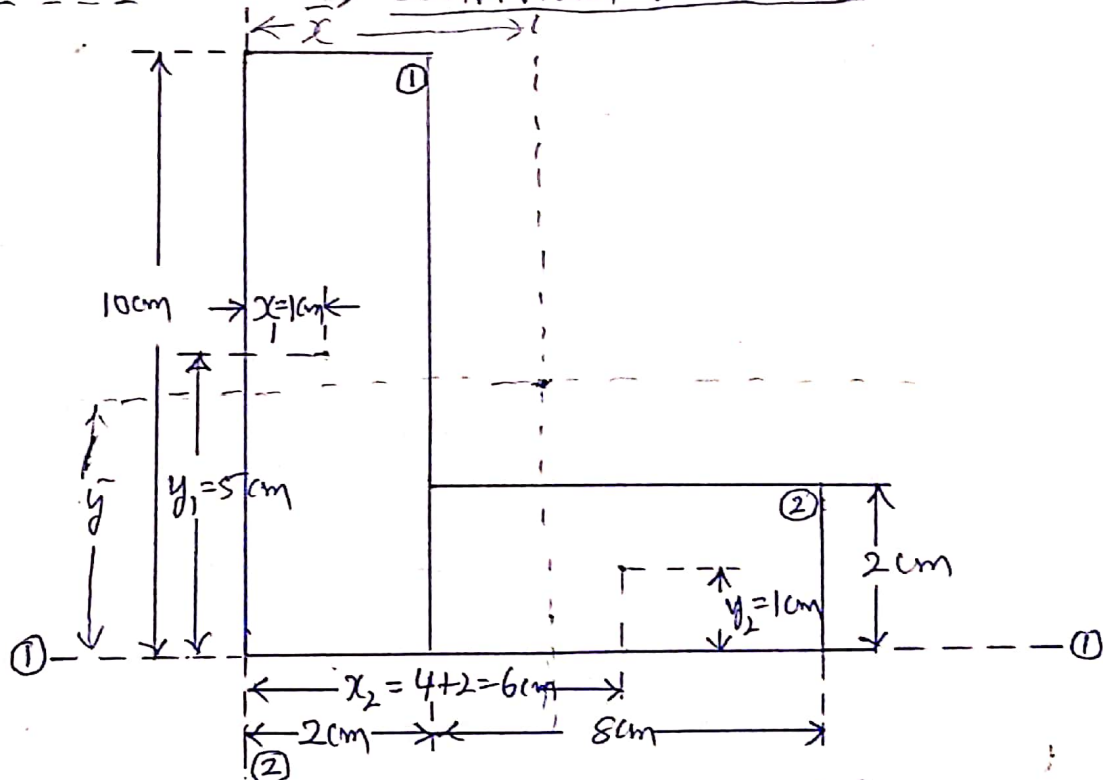
4) Determine the M.I and radius of gyration about centroidal xx, centroidal yy and zz-axis for the 'L' section shown in fig



Solution:

(2)

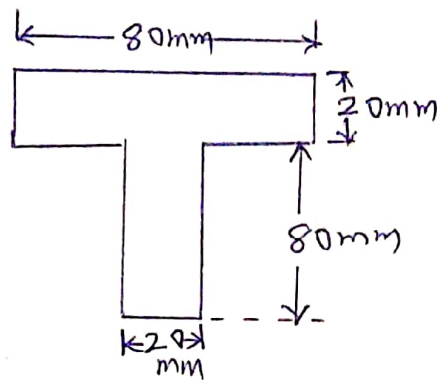
i) Centroidal distance



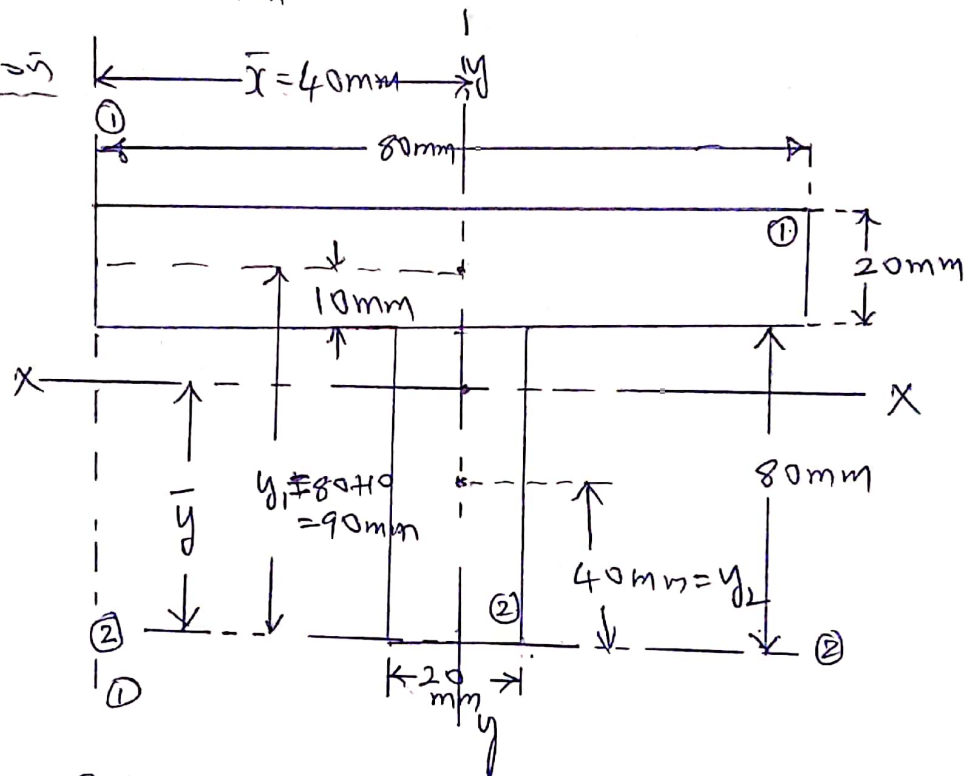
$$\bar{x} = \frac{a_1 x_1 + a_2 x_2}{a_1 + a_2} = \frac{(2 \times 10)(1) + 8 \times 2(6)}{20 + 16} = 3.22 \text{ cm}$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2}{a_1 + a_2} = \frac{(2 \times 10) \times 5 + 8 \times 2(1)}{20 + 16} = \underline{\underline{3.22 \text{ cm}}}$$

5) Determine the M.I of T-section about centroidal xx , yy and zz axis. Also determine radii of gyration.



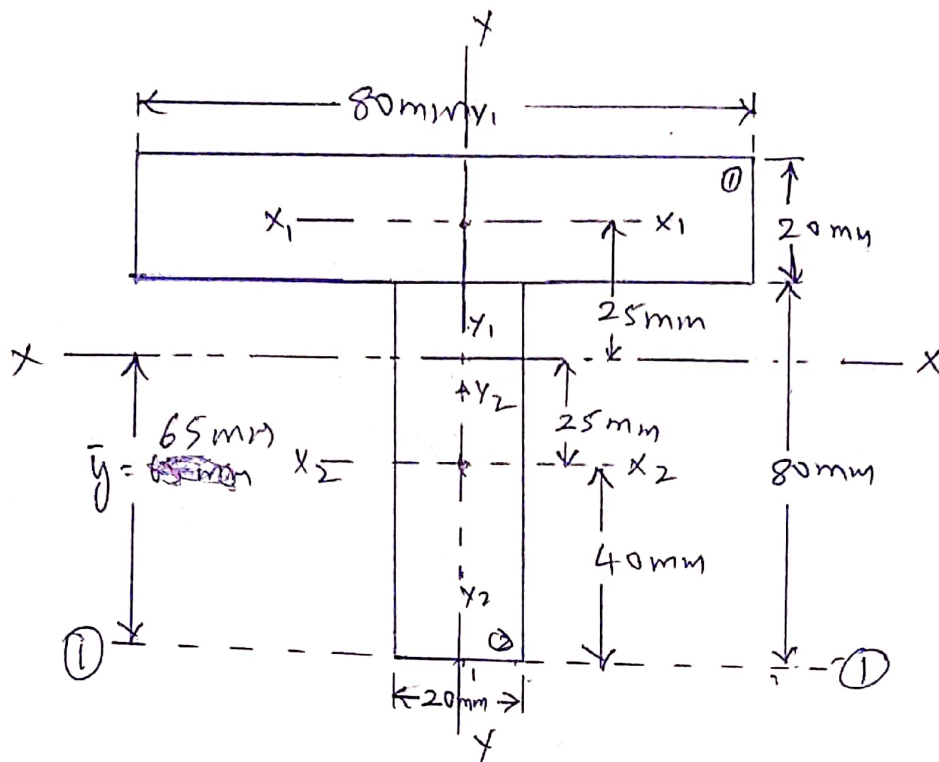
Solution



$$\bar{y} = \frac{a_1 y_1 + a_2 y_2}{a_1 + a_2}$$

$$= \frac{(80 \times 20) \times 90 + (80 \times 20) \times 40}{1600 + 1600} = \underline{65\text{mm}}$$

$$\bar{y} = 65\text{mm from axis (2)(2).}$$



$$I_{xx} = \left[\left(\frac{80 \times 20^3}{12} \right) + 80 \times 20 (25)^2 \right] + \left[\left(\frac{20 \times 80^3}{12} \right) + 20 \times 80 (25)^2 \right]$$

$$= 2906666.67 \text{ mm}^4$$

$$I_{yy} = \left[\frac{20 (80)^3}{12} + 20 \times 80 (0)^2 \right] + \left[\frac{80 \times 20^3}{12} + 80 \times 20 (0)^2 \right]$$

$$= 893333.33 \text{ mm}^4$$

Radius of gyration,

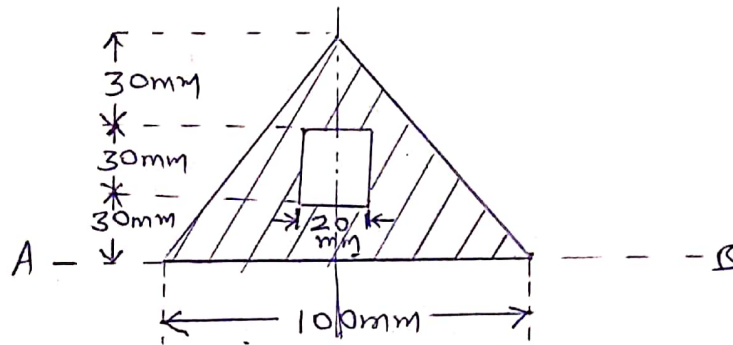
$$K_x = \sqrt{\frac{I_{xx}}{A}} = \sqrt{\frac{2906666.67}{80 \times 20 + 80 \times 20}} = 9.53 \text{ mm}$$

$$K_y = \sqrt{\frac{I_{yy}}{A}} = \sqrt{\frac{893333.33}{80 \times 20 + 80 \times 20}} = 16.71 \text{ mm}$$

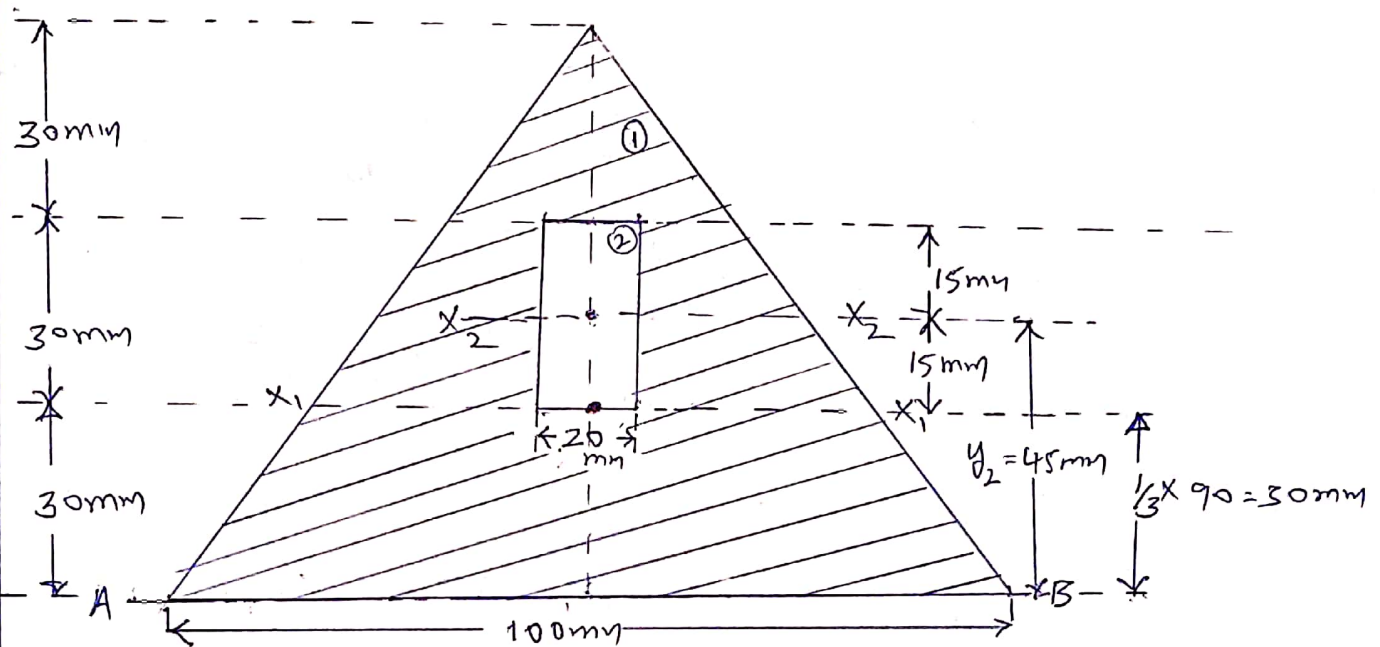
$$K_z = \sqrt{\frac{I_{xx} + I_{yy}}{A}} = \sqrt{\frac{2906666.67 + 893333.33}{80 \times 20 + 80 \times 20}}$$

$$= 34.46 \text{ mm}$$

6) Determine the radius of gyration of shaded area shown in fig. about the axis AB.



Solution:-



$$I_{AB} = \left(\frac{100(90)^3}{36} + \frac{1}{2} \times 100 \times 90 (30)^2 \right) + \left[\frac{20 \times 30^3}{12} + 20 \times 30 (45)^2 \right]$$

$$= \underline{\underline{4815000 \text{ mm}^4}}$$

$$A = \frac{1}{2} \times 100 \times 90 - 20 \times 30 = 3900 \text{ mm}^2$$

Radius of gyration about axis AB,

$$K_{AB} = \sqrt{\frac{I_{AB}}{A}} = \sqrt{\frac{4815000}{3900}}$$

$$= \underline{\underline{35.14 \text{ mm}}}$$